

SCENE 63: THE NHL ARSENAL — Master Treatment & Novel Therapies



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1. Chemo weapon rack

WHAT YOU SEE

'THE CHEMO WEAPON RACK' shelf

WHAT IT ENCODES

The unifying principle of B-cell NHL therapy is immunochemotherapy: an anti-CD20 antibody (RITUXIMAB, the 'R') added to a cytotoxic backbone, with regimen INTENSITY scaling to tumor aggressiveness. Aggressive B-cell disease (DLBCL) gets R-CHOP; indolent disease (follicular, marginal zone) gets gentler bendamustine-rituximab or watchful waiting; very high-grade disease (Burkitt, double-hit) gets dose-intensive DA-EPOCH-R or hyper-CVAD with CNS coverage. Across the board, screen HEPATITIS B before rituximab and give tumor-lysis prophylaxis for bulky/high-grade tumors.

BOARD TRAP

WRONG to use the same regimen for every B-cell lymphoma. The discriminator: match intensity to biology — indolent (watch-and-wait or BR), aggressive (R-CHOP), very high-grade Burkitt/double-hit (DA-EPOCH-R, never plain R-CHOP) — and remember T-cell lymphomas are CD20-NEGATIVE, so the 'R' is omitted entirely.

2. R-CHOP (top rack)

WHAT YOU SEE

'R-CHOP 21' rifle, top rack

WHAT IT ENCODES

R-CHOP = Rituximab + Cyclophosphamide + Hydroxydaunorubicin (DOXORUBICIN) + Oncovin (VINCRIStINE) + Prednisone, given every 21 days for ~6 cycles; it is the standard frontline for DLBCL and most aggressive B-cell NHL and CURES ~60% of DLBCL. Key toxicities map to each drug: doxorubicin → CARDIOTOXICITY (dose-dependent; check baseline ejection fraction), vincristine → peripheral NEUROPATHY and constipation, cyclophosphamide → hemorrhagic cystitis/myelosuppression, prednisone → hyperglycemia/mood. The newer pola-R-CHP (polatuzumab replacing vincristine) modestly improved outcomes in POLARIX.

BOARD TRAP

WRONG to give R-CHOP to Burkitt or double-hit lymphoma. The discriminator: R-CHOP is INADEQUATE for the highest-grade tumors (Burkitt, high-grade B-cell with MYC+BCL2 rearrangements) — they need dose-intensive, CNS-penetrating regimens (DA-EPOCH-R, hyper-CVAD, CODOX-M/IVAC); using R-CHOP for Burkitt is a classic exam misstep that undertreats a curable disease.

3. Indolent / BR

WHAT YOU SEE

'FL/INDOLENT BR' label

WHAT IT ENCODES

Bendamustine + Rituximab (BR) is a standard frontline for symptomatic INDOLENT B-cell NHL — follicular, marginal-zone, and (in older/less-fit patients) mantle cell — and was non-inferior/better tolerated than R-CHOP in the StiL and BRIGHT trials. Bendamustine is a hybrid alkylator/purine-analog with notable LYMPHOPENIA and infection/opportunistic-infection risk (PJP, herpesvirus reactivation), so consider antimicrobial prophylaxis. Asymptomatic, low-tumor-burden follicular lymphoma is often just OBSERVED ('watch and wait') rather than treated.

BOARD TRAP

WRONG to rush to treat asymptomatic low-burden follicular lymphoma. The discriminator: follicular lymphoma is indolent and INCURABLE, and early treatment of asymptomatic low-burden disease does not improve survival (GELF criteria define when to treat) — so 'watch and wait' is the correct answer unless there is bulky disease, cytopenias, symptoms, or organ compromise.

4. DA-EPOCH-R (middle)

WHAT YOU SEE

'DA-EPOCH-R' weapon, middle rack

WHAT IT ENCODES

Dose-Adjusted EPOCH-R (Etoposide, Prednisone, Oncovin/vincristine, Cyclophosphamide, Hydroxydaunorubicin/doxorubicin + Rituximab) uses continuous-infusion dosing that is escalated by nadir counts; it is preferred for BURKITT lymphoma, HIGH-GRADE/DOUBLE-HIT B-cell lymphoma, and primary mediastinal large B-cell lymphoma. It delivers more dose intensity and better CNS coverage than R-CHOP. Because Burkitt's proliferation is near 100%, aggressive TUMOR-LYSIS prophylaxis (hydration, rasburicase/allopurinol) and intrathecal CNS prophylaxis are mandatory.

BOARD TRAP

WRONG to omit tumor-lysis and CNS prophylaxis when escalating to DA-EPOCH-R for Burkitt/high-grade disease. The discriminator: the same explosive proliferation that demands DA-EPOCH-R also creates the highest TLS risk and CNS-relapse risk — so rasburicase (not just allopurinol) for high-risk TLS plus intrathecal methotrexate/cytarabine are part of the package, not optional add-ons.

5. Double-hit

WHAT YOU SEE

'BL/DOUBLE-HIT' tag

WHAT IT ENCODES

'Double-hit' (high-grade B-cell lymphoma with MYC + BCL2 and/or BCL6 REARRANGEMENTS on FISH) is biologically near-Burkitt in aggressiveness and behaves poorly with standard R-CHOP, so it is treated like Burkitt with intensive therapy (DA-EPOCH-R) plus CNS prophylaxis. It is distinct from a 'double-EXPRESSOR' lymphoma (MYC + BCL2 overexpressed by IHC without rearrangement), which is prognostically worse but still treated with R-CHOP. Every aggressive GCB-type DLBCL with high Ki-67 should be sent for MYC/BCL2/BCL6 FISH.

BOARD TRAP

WRONG to treat double-hit lymphoma as ordinary DLBCL with R-CHOP. The discriminator: confirm REARRANGEMENTS by FISH (genetic, not IHC) — double-hit requires DA-EPOCH-R-type intensification and CNS prophylaxis, whereas a mere double-expressor (protein by IHC) gets standard R-CHOP; conflating expressor with hit is a frequent trap.

6. Legacy / R-CVP (bottom)

WHAT YOU SEE

'LEGACY' rifle, 'R-CVP'

WHAT IT ENCODES

R-CVP = Rituximab + Cyclophosphamide + Vincristine + Prednisone — essentially R-CHOP WITHOUT the anthracycline (no doxorubicin) — used as a gentler regimen for FRAIL/elderly patients or low-grade disease where cardiotoxicity must be avoided (e.g., reduced ejection fraction, prior anthracycline exposure). Dropping doxorubicin spares the heart but lowers efficacy in aggressive disease, so R-CVP is not the choice for fit DLBCL patients.

BOARD TRAP

WRONG to substitute R-CVP for R-CHOP in a fit DLBCL patient just to avoid toxicity. The discriminator: the ANTHRACYCLINE (doxorubicin) is the curative backbone in aggressive lymphoma — omitting it (R-CVP) sacrifices cure rates, so reserve R-CVP for genuinely anthracycline-ineligible patients (poor cardiac function); fit DLBCL patients should get full R-CHOP with cardiac monitoring.

7. Rituximab mechanism

WHAT YOU SEE

'RITUXIMAB MECHANISM' antibody on B-cell

WHAT IT ENCODES

Rituximab is a chimeric anti-CD20 IgG1 monoclonal antibody that depletes B-cells via four mechanisms: antibody-dependent cellular cytotoxicity (ADCC), complement-dependent cytotoxicity (CDC), antibody-dependent phagocytosis, and direct apoptosis signaling. Because CD20 is expressed on most B-cell NHLs but NOT on plasma cells or stem cells, normal B-cell recovery eventually occurs. Mandatory pre-treatment step: screen HEPATITIS B (HBsAg + anti-HBc) and give antiviral prophylaxis (entecavir/tenofovir) for positives, because rituximab can cause fatal HBV REACTIVATION; also a rare risk of PML (JC virus).

BOARD TRAP

WRONG to start rituximab without hepatitis B serologies. The discriminator: rituximab-induced B-cell depletion can REACTIVATE latent HBV (even in anti-HBc-positive, HBsAg-negative 'resolved' infection) leading to fulminant hepatitis — so HBsAg AND anti-HBc must be checked and prophylaxis started before the first dose; skipping this is a classic patient-safety exam trap.

8. CD20 target

WHAT YOU SEE

'CD20' on the B-cell with mechanisms (CDC/ADCC/PHAGO/APOP)

WHAT IT ENCODES

CD20 is a B-lineage-restricted surface phosphoprotein expressed from the pre-B stage through mature B-cells but lost on plasma cells and absent on stem cells — making it the canonical immunotherapy target in B-cell NHL. OBINUTUZUMAB is a glycoengineered, type-II humanized anti-CD20 with enhanced ADCC and direct cell death, used in follicular lymphoma and CLL. CD20-targeting is why T-cell lymphomas (CD20-negative) get NO benefit from rituximab.

BOARD TRAP

WRONG to expect rituximab/anti-CD20 to work in a T-cell lymphoma or a CD20-negative tumor. The discriminator: CD20 is a B-CELL marker — T-cell lymphomas (PTCL, ALCL, ATLL, mycosis fungoides) are CD20-negative, and even some B-cell tumors lose CD20 after prior rituximab; verify CD20 expression by flow/IHC before assuming the antibody will help.

9. IT MTX / CNS prophylaxis

WHAT YOU SEE

'IT MTX' syringe to brain, 'CNS PROPHYLAXIS' lighthouse

WHAT IT ENCODES

CNS prophylaxis aims to prevent leptomeningeal/parenchymal relapse in lymphomas at high CNS risk, delivered as INTRATHECAL methotrexate/cytarabine and/or HIGH-DOSE systemic methotrexate (which crosses the blood-brain barrier). It is indicated for Burkitt, lymphoblastic lymphoma, high-grade/double-hit lymphoma, testicular DLBCL, and DLBCL with high CNS-IPI or involvement of high-risk sites (e.g., paranasal sinus, epidural, breast, bone marrow). High-dose methotrexate requires leucovorin rescue, urine alkalinization, and methotrexate-level monitoring.

BOARD TRAP

WRONG to give standard intravenous methotrexate doses and expect CNS coverage, or to forget leucovorin rescue. The discriminator: only HIGH-DOSE (≥ 3 g/m²) methotrexate penetrates the CNS, and it mandates leucovorin rescue + urinary alkalinization + drug-level monitoring (and avoiding interacting drugs like NSAIDs, PPIs, trimethoprim) to prevent fatal toxicity; intrathecal delivery is the alternative route.

10. CNS-IPI rules board

WHAT YOU SEE

'RULES BOARD — CNS protection must be specific'

WHAT IT ENCODES

The CNS-IPI is a 6-point risk score for CNS relapse in DLBCL combining the 5 IPI factors (age >60, elevated LDH, ECOG performance status >1, stage III/IV, >1 extranodal site) PLUS adrenal/KIDNEY involvement. Scores stratify low (0-1, ~0.6%), intermediate (2-3, ~3.4%), and high (4-6, ~10%) CNS-relapse risk, identifying which DLBCL patients warrant CNS-directed prophylaxis. Certain anatomic sites (testis, breast, epidural, paranasal sinus) independently raise risk regardless of score.

BOARD TRAP

WRONG to give CNS prophylaxis to every DLBCL patient or to omit it based on score alone in a high-risk SITE. The discriminator: routine CNS prophylaxis is reserved for HIGH CNS-IPI (4-6) and specific high-risk anatomic sites (testicular, breast, epidural) — over-prophylaxis exposes low-risk patients to methotrexate toxicity, while ignoring testicular involvement (which also needs scrotal radiation) undertreats a sanctuary site.

11. CAR-T launch bay CD19

WHAT YOU SEE

'CAR-T LAUNCH BAY — CD19'

WHAT IT ENCODES

Chimeric antigen receptor (CAR) T-cell therapy genetically engineers the patient's own T-cells to express an anti-CD19 receptor; approved products for relapsed/refractory aggressive B-cell lymphoma are axicabtagene ciloleucel (axi-cel), tisagenlecleucel (tisa-cel), and lisocabtagene maraleucel (liso-cel); BREXUcabtagene autoleucel (brexu-cel) is for relapsed mantle cell lymphoma. Signature toxicities are CYTOKINE RELEASE SYNDROME (fever/hypotension/hypoxia, treated with tocilizumab ± steroids) and ICANS (immune-effector-cell-associated neurotoxicity, treated with steroids).

BOARD TRAP

WRONG to treat CAR-T cytokine release syndrome the same as ICANS. The discriminator: CRS (IL-6-driven fever/hypotension) responds to TOCILIZUMAB (anti-IL-6R) ± steroids, whereas ICANS (neurotoxicity — aphasia, encephalopathy, seizures) does NOT respond well to tocilizumab and is managed with CORTICOSTEROIDS — and tocilizumab can paradoxically worsen ICANS by raising CNS IL-6; knowing which drug for which syndrome is high-yield.

12. CAR-T bays/lines

WHAT YOU SEE

'BAY 1/2/3 — 3RD/2ND/1ST LINE, MCL'

WHAT IT ENCODES

CD19 CAR-T therapy has moved EARLIER in DLBCL: based on ZUMA-7 (axi-cel) and TRANSFORM (liso-cel), it is now SECOND-LINE for primary-refractory or early-relapsing (within 12 months) DLBCL, outperforming the old standard of salvage chemo + autologous stem-cell transplant. Patients relapsing LATER (>12 months) may still go to salvage chemo/autoSCT first. In mantle cell lymphoma, brexu-cel CAR-T is used after BTK-inhibitor failure.

BOARD TRAP

WRONG to default every relapsed DLBCL to salvage chemo + autologous transplant. The discriminator: for PRIMARY-REFRACTORY or EARLY-relapse (≤ 12 months) DLBCL, CAR-T is now preferred second-line (ZUMA-7/TRANSFORM), whereas autologous transplant after salvage chemo remains appropriate for LATE relapses that are still chemosensitive — timing of relapse decides the pathway.

13. Bispecifics (CD20xCD3)

WHAT YOU SEE

'BISPECIFICS — CD20 / CD3' biotech wing

WHAT IT ENCODES

Bispecific T-cell engagers (e.g., GLOFITAMAB, EPCORITAMAB, mosunetuzumab) are antibodies with one arm binding tumor CD20 and the other binding CD3 on T-cells, physically bridging T-cells to the tumor to trigger killing. They are 'OFF-THE-SHELF' (no manufacturing lag, unlike CAR-T) and used in relapsed/refractory follicular and large B-cell lymphoma. Like CAR-T they cause CYTOKINE RELEASE SYNDROME, mitigated by step-up dosing and prophylaxis; mosunetuzumab is approved for relapsed follicular lymphoma.

BOARD TRAP

WRONG to assume bispecifics carry the same logistics as CAR-T. The discriminator: bispecific antibodies are OFF-THE-SHELF (immediately available, given as a drug with step-up dosing), whereas CAR-T requires apheresis, weeks of manufacturing, and lymphodepletion — making bispecifics an option for patients who cannot wait for or are ineligible for CAR-T, though both share CRS risk.

14. ADC concept

WHAT YOU SEE

'ADC — different target, same cell'

WHAT IT ENCODES

Antibody-drug conjugates couple a tumor-antigen-targeting antibody to a cytotoxic payload, delivering chemo selectively to lymphoma cells. POLATUZUMAB VEDOTIN targets CD79b (with MMAE payload) and is used in pola-R-CHP frontline (POLARIX) and in relapsed DLBCL; LONCASTUXIMAB tesirine targets CD19 (with a PBD-dimer payload) for relapsed/refractory DLBCL. ADCs let a single B-cell express several druggable antigens (CD19, CD20, CD79b, CD30), enabling sequential targeting.

BOARD TRAP

WRONG to assume an ADC's toxicity comes from the antibody. The discriminator: ADC toxicity tracks the CYTOTOXIC PAYLOAD — polatuzumab's MMAE causes peripheral NEUROPATHY (like vincristine/brentuximab), so it's combined with R-CHP minus vincristine to avoid stacking neurotoxicity; recognizing the payload class predicts the side-effect profile.

15. Brentuximab (BRENT)

WHAT YOU SEE

'BRENT — CD30 brentuximab' arrow

WHAT IT ENCODES

Brentuximab vedotin is an anti-CD30 antibody-drug conjugate carrying the antimetabolic MMAE payload; it targets CD30+ tumors — frontline (with CHP) for CD30+ peripheral T-cell lymphomas including ALCL (ECHELON-2), and used in Hodgkin lymphoma (with AVD frontline per ECHELON-1, and in relapse). Its hallmark dose-limiting toxicity is peripheral SENSORY NEUROPATHY from the MMAE; combining it with bleomycin is avoided due to pulmonary toxicity.

BOARD TRAP

WRONG to add brentuximab on top of a full vincristine-containing regimen, or to give it with bleomycin. The discriminator: brentuximab's MMAE payload causes neuropathy, so in PTCL it REPLACES vincristine (CHOP→CHP), and in Hodgkin frontline it replaces bleomycin (ABVD→AVD) to avoid additive lung toxicity — matching the ADC to the regimen it modifies is the high-yield point.